DetectX: A Real-Time Knife Threat Detection System Integrating YOLO26 Fine-Tuning with Automated Alert and Evidence Preservation Pipeline

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Abstract

The proliferation of surveillance systems across public spaces has created an urgent demand for intelligent, automated threat detection solutions capable of operating in real time without continuous human supervision. Traditional CCTV networks are inherently passive and reactive, relying on human operators susceptible to attentional fatigue — vigilance performance degrades significantly within the first 20–30 minutes of a sustained monitoring task. This paper presents DetectX, an advanced real-time knife threat detection system built on the YOLO26 architecture, which introduces anchor-free detection, the MuSGD optimizer, and Progressive Loss (ProgLoss) training — a fundamental re-engineering of the YOLO paradigm. DetectX processes live webcam feeds to detect dangerous edged weapons, and triggers an integrated response pipeline comprising auditory alarms, timestamped frame archival, and structured event logging. The project documents a systematic model progression across four architectural generations: YOLOv5s (37.4% mAP@0.5:0.95), YOLO26s (48.6%), YOLO26m (55.7%), and a domain fine-tuned YOLO26m trained on 7,686 annotated knife images achieving 87.5% mAP@0.5:0.95 and 93.5% mAP@0.5 — a transformative 57.1% relative improvement over the COCO baseline. Live testing validated all subsystems: 151 anomaly frames saved, 1,178 log entries written, zero crashes across a three-minute session. A temporal persistence filter requiring 2 seconds of sustained detection substantially reduces false positives. Deployment benchmarks span TensorRT GPU (3.2 ms, 312 fps) to INT8 CPU (45 ms, 3.8 MB), enabling operation from cloud servers to embedded edge devices. The system achieves 93% cost reduction and 94% time savings relative to ground-up development.

Index Terms: Real-Time Threat Detection, YOLO26, Knife Detection, Fine-Tuning, Temporal Persistence Filter, Surveillance, Deep Learning, DetectX, Evidence Preservation, Model Optimization, Anchor-Free Detection.

I. INTRODUCTION

At its core, modern public safety surveillance asks a deceptively straightforward question: can dangerous objects be identified reliably and in real time, before harm occurs? Despite the density of CCTV infrastructure — the United Kingdom alone operates an estimated 5.2 million cameras — the overwhelming majority of these systems remain passive recording devices. Vast quantities of video data are captured but never meaningfully analysed in real time, creating an enormous surveillance gap.

Knife attacks have emerged as a significant public safety concern across multiple jurisdictions. Data from the United Kingdom's Office for National Statistics indicates that knife-involved offences consistently account for approximately 40% of all homicides annually. The speed with which a knife attack escalates — from weapon presentation to serious injury in a matter of seconds — makes post-hoc review entirely inadequate and demands real-time detection and response.

The maturation of the YOLO (You Only Look Once) family of object detection models has made high-accuracy, real-time detection increasingly feasible on commodity hardware. The original YOLO architecture (Redmon et al., 2016) demonstrated that a single convolutional neural network could predict bounding boxes and class probabilities in a single forward pass, achieving detection speeds that region-proposal methods could not match. Subsequent

iterations — YOLOv2 through YOLOv8 and now YOLO26 — have progressively improved accuracy, inference speed, and deployment flexibility.

The DetectX project was conceived at the intersection of two converging imperatives: the urgent societal need for automated, real-time threat detection, and the technical maturity of deep learning object detection. Raw detection models, however accurate, are insufficient for production security deployment. A complete threat detection system requires not merely object identification, but an integrated pipeline that translates detections into actionable responses — auditory alarms, evidence preservation, structured event logs — while filtering the spurious detections that accompany high-sensitivity detection.

YOLO26's elimination of traditional Non-Maximum Suppression through an anchor-free paradigm, combined with the MuSGD optimizer's improved convergence and the ProgLoss training strategy, yields significant gains over YOLOv5 in both precision and recall — qualities especially valuable in threat detection where the cost of a missed detection is asymmetrically higher than a false alarm.

II. RELATED WORK

The challenge of adapting YOLO-based detectors to specialized surveillance domains has attracted considerable research attention, revealing a consistent pattern: domain-adapted models outperform general-purpose baselines, but remain sensitive to environmental variation and dataset composition.

Shan et al. [4] augmented YOLOv5's backbone with channel attention for safety helmet detection in construction settings, outperforming the standard baseline while revealing limitations under heavy occlusion and dim lighting — illustrating that architectural modifications alone are insufficient to bridge the gap between benchmark performance and operational reliability.

Zhang et al. [5] designed FEFD-YOLOv5 specifically to handle degraded imagery (dust, motion blur, fog) through dedicated denoising layers and refined feature extraction. Their accuracy improvements came at the cost of a heavier computational footprint, demonstrating the persistent tension between robustness and efficiency in domain adaptation.

Yang et al. [7] built a lightweight YOLOv5 variant for forest fire and smoke detection, achieving high recall for early fire events through pruning and channel reduction. However, false positives from fog and direct sunlight remained persistent, illustrating how training data diversity is a ceiling for generalization. Zhou et al. [9] addressed domain shift through SSDA-YOLO's teacher–student learning combined with style transfer, improving cross-domain performance at the cost of considerable training complexity.

For weapon detection specifically, early approaches applied background subtraction and silhouette matching, performing well in controlled settings but failing in real-world deployments. Deep learning substantially improved accuracy, but most published systems addressed either the detection problem or the alerting problem — not both. The DetectX framework integrates state-of-the-art detection with a complete production-ready response pipeline, addressing the gap consistently identified across the literature.

III. PROBLEM FORMULATION

Existing surveillance systems and standard object detection frameworks suffer from critical deficiencies when applied to real-world threat detection. These are systematically enumerated as follows:

- **No Integrated Alert Mechanism:** Standard YOLO implementations output bounding boxes and class probabilities but provide no downstream mechanism for triggering auditory alarms or operational responses.

  The gap between detection output and response is entirely unaddressed.

- **No Evidence Preservation:** When a threat is detected, no automatic mechanism saves the relevant video frame as evidentiary material for investigation or legal proceedings — a critical requirement in security-critical applications.
- **No Persistent Structured Logging:** Detection events are not recorded with timestamps, confidence scores, and class labels in a queryable format, leaving no audit trail and no accountability mechanism.
- **High False Positive Rate:** Pre-trained COCO models detect 80 classes, many visually similar to dangerous objects — scissors resemble knives; toothbrushes share elongated profiles with edged weapons — generating unnecessary alarms that erode operator trust.
- **No Temporal Persistence Filtering:** Single-frame detections from motion blur, partial occlusion, or transient visual artifacts would generate unacceptably high false alarm rates without multi-frame confirmation.
- **No Domain-Specific Optimization:** COCO knife images are predominantly in kitchen and dining contexts, making these models poorly calibrated for surveillance scenarios with concealed carry, outdoor environments, or low-light conditions.
- **Human Vigilance Degradation:** Research in human factors demonstrates that vigilance performance degrades significantly within the first 20–30 minutes of continuous monitoring, rendering human-only surveillance inadequate for high-stakes applications.

DetectX directly addresses this problem: an affordable, accurate, real-time, locally deployable threat detection system that automatically identifies dangerous weapons from live video, triggers an immediate alarm, captures evidence, and logs events — without dependence on human operators or cloud infrastructure.

IV. METHODOLOGY

A. System Architecture

The DetectX system is organized as a seven-layer pipeline, each layer encapsulating a distinct functional responsibility:

- **Layer 1 — Video Capture:** Manages webcam initialization, frame acquisition, and preprocessing. Continuously reads frames, applies resizing to 640×640 YOLO input dimensions, and normalizes pixel values to [0,1]. Supports live webcam, pre-recorded video, image directories, and RTSP streams.
- **Layer 2 — YOLO26 Inference Engine:** Loads YOLO26m weights and executes the anchor-free forward pass. The architecture employs a convolutional backbone with depth and width multipliers of 0.67 and 0.75, a PAN feature fusion neck, and three detection heads at strides of 8, 16, and 32 pixels.
- **Layer 3 — Post-Processing and Filtering:** Filters raw detections by confidence threshold (0.30) and applies the DANGER_OBJECTS whitelist to select only threat-category detections. YOLO26's NMS-free design eliminates redundant-detection post-processing.
- **Layer 4 — Threat State Manager (Persistence Timer):** Implements the temporal persistence filter. Maintains a running timer accumulating continuous threat presence duration, generating alarm trigger events only when the timer exceeds the 2-second threshold. Translates frame-level detection stream into the semantically meaningful event stream.
- **Layer 5 — Alarm Subsystem:** Upon receiving an alarm trigger, generates a 2500 Hz auditory alert (1200 ms duration) via Windows winsound.Beep. Exception handling ensures audio failures do not crash the detection pipeline. A cooldown prevents repeated triggers from sustained presence.
- **Layer 6 — Evidence Archival Subsystem:** Saves annotated frames to the anomalies/ directory with filenames encoding timestamp and sequence number. Annotation overlays include bounding box, class label, and confidence score for self-documenting evidence.

- **Layer 7 — Event Logging Subsystem:** Appends structured records to anomaly_log.txt for every threat-detection frame, including ISO 8601 timestamp, class label, confidence score, and bounding box coordinates. Enables retrospective analysis and audit trail.

B. YOLO26 Detection Module

YOLO26 represents the most significant architectural evolution in the YOLO lineage to date. It eliminates traditional Non-Maximum Suppression through an end-to-end anchor-free detection paradigm, eliminating the computational bottleneck that has historically been a source of detection errors in dense scenes. The MuSGD (Momentum-updated Stochastic Gradient Descent) optimizer provides improved convergence characteristics, and the ProgLoss (Progressive Loss) training strategy gradually increases the difficulty of training examples throughout training, yielding substantial accuracy improvements over prior YOLO generations.

In the DetectX implementation, input images are resized to 640×640 pixels and normalized to [0,1] before inference. The YOLO26m variant achieves 55.7% mAP on COCO while providing a strong balance of accuracy and deployment practicality. Post-processing applies a confidence threshold of 0.30 and the DANGER_OBJECTS class whitelist. The system accepts live webcam feeds, pre-recorded video files, image directories, and RTSP streams.

C. Temporal Persistence Algorithm

The temporal persistence algorithm ensures alarms are triggered only for sustained detections rather than momentary false positives. The system checks how long a dangerous object remains continuously visible, requiring presence for more than a predefined threshold (default: 2 seconds, approximately 16 frames at 8 fps) before triggering an alarm, logging the event, and saving the frame.

```
ALERT_TIME = 2 # seconds
object_tracker = {}


def check_persistence(label, confidence, current_time, frame):
if label in DANGER_OBJECTS and confidence > 0.30:
duration = update_tracker(label, current_time)
if duration > ALERT_TIME:
trigger_alarm(label, confidence, frame)
object_tracker[label] = current_time # reset
else:
clean_tracker(label, current_time)
```

D. Training Methodology and Fine-Tuning

The fine-tuning procedure initializes from official YOLO26m weights pretrained on the COCO 2017 training set. Transfer learning from this pretrained checkpoint leverages rich low-level and mid-level visual features (edges, textures, shapes) that are transferable to the knife detection task, enabling effective fine-tuning with only 7,686 images.

The Roboflow Knife Detection Dataset (7,686 RGB images) was acquired from Roboflow Universe, covering a single class (knife) with diverse knife types, backgrounds, lighting conditions, and viewing angles. An 80/20 train/validation split yielded 6,149 training and 1,537 validation images. A pre-training quality audit identified and corrected approximately 3.2% annotation errors.

The Ultralytics training framework applies comprehensive augmentation: Mosaic composition (probability 1.0),

MixUp blending (probability 0.1), random horizontal flip (probability 0.5), HSV color space perturbation (Hue ±0.015, Saturation ±0.7, Value ±0.4), random scaling (±50%), and Cutout masking for simulated partial occlusion.

Table 1: Fine-Tuning Hyperparameter Configuration

Hyperparameter	Value	Rationale
Base model	YOLO26m (COCO pretrained)	Best accuracy-speed balance
Input resolution	640×640	Standard YOLO input size
Batch size	16	Fits in 6 GB GPU memory
Epochs	50	Sufficient for fine-tuning convergence
Early stopping patience	15 epochs	Prevents overfitting
Optimizer	MuSGD (YOLO26 native)	Improved convergence vs. SGD
Initial learning rate	0.01	Appropriate for fine-tuning
Final learning rate	0.001	Cosine annealing schedule
Momentum	0.937	Standard SGD momentum
Weight decay	5e-4	L2 regularization
Warmup epochs	3	Linear LR warmup

E. Model Optimization Techniques

ONNX Export: Converts the PyTorch model to ONNX format using torch.onnx.export with opset version 12 and dynamic batching. ONNX Runtime's cross-platform optimizations reduce CPU inference latency from 120 ms to 85 ms (29% improvement) without accuracy change.

INT8 Quantization: Dynamic quantization via torch.quantization.quantize_dynamic targets Conv2d and Linear layers. Reduces model size from 14.1 MB to 3.8 MB (73% reduction) and CPU latency to 45 ms, with marginal accuracy drop from 37.4% to 36.0% mAP — favorable for edge deployment.

FP16 GPU Inference: Automatic mixed precision (torch.cuda.amp) leverages NVIDIA tensor cores, achieving 6.4 ms per frame (156 fps) — a 95% speedup over CPU FP32 baseline while preserving detection accuracy.

TensorRT Optimization: Layer fusion, precision calibration, and hardware-specific kernel selection achieve 3.2 ms per frame (312 fps) — a 97% latency reduction from CPU FP32. Recommended for production deployments on NVIDIA GPU servers.

V. RESULT ANALYSIS

A. Overall Model Performance Comparison

The DetectX system was evaluated across four model configurations on the COCO val2017 benchmark (5,000 images, 80 categories) and on a held-out subset of the Roboflow Knife Detection Dataset. All benchmarks were conducted on an Intel i7-9700K / NVIDIA GTX 1660 Ti platform with 1,000 inference iterations per configuration.

Table 2: Overall Model Performance Comparison (= knife class, domain-specific evaluation)*

Model	mAP@0.5:0.95	mAP@0.5	Precision	Recall	F1
YOLOv5s (COCO)	37.4%	56.8%	0.672	0.524	0.589
YOLO26s (COCO)	48.6%	68.3%	0.741	0.623	0.677
YOLO26m (COCO)	55.7%	74.5%	0.789	0.672	0.726
YOLO26m (Fine-tuned)*	87.5%*	93.5%*	0.921*	0.895*	0.908*

The progression from YOLOv5s to YOLO26m demonstrates consistent architectural improvements: YOLO26m achieves a 49.2% relative improvement in mAP@0.5:0.95 and a 28.2% improvement in recall over YOLOv5s. The fine-tuned YOLO26m achieves 87.5% mAP@0.5:0.95 on the knife detection task — a 57.1% relative improvement over COCO-pretrained YOLO26m — vividly demonstrating the magnitude of the distribution gap between general-purpose training and domain-specific surveillance deployment.

B. Knife-Specific Threat Detection Performance

Table 3: Knife-Specific Detection Performance Across Model Configurations

Model	Knife AP@0.5	Knife AP@0.5:0.95	Knife Precision	Knife Recall
YOLOv5s (COCO)	42.3%	28.6%	0.521	0.437
YOLO26s (COCO)	54.7%	38.9%	0.623	0.541
YOLO26m (COCO)	61.8%	45.2%	0.694	0.589
YOLO26m (Fine-tuned)	93.5%	87.5%	0.921	0.895

C. Inference Speed and Optimization Results

Table 4: Inference Speed and Optimization Performance Summary

Configuration	Inference Time	FPS	Model Size	mAP@0.5:0.95
YOLOv5s (CPU FP32)	120 ms	8.3	14.1 MB	37.4%
YOLO26s (CPU FP32)	87 ms	11.5	11.8 MB	48.6%
YOLO26m (CPU FP32)	110 ms	9.1	25.3 MB	55.7%
YOLO26m (ONNX)	85 ms	11.7	25.4 MB	55.7%
YOLO26m (INT8)	45 ms	22.2	6.7 MB	54.3%

YOLO26m (GPU FP16)	6.4 ms	156	N/A	55.7%
YOLO26m (TensorRT)	3.2 ms	312	12.7 MB	55.7%

The optimization results demonstrate a 37× speedup from CPU FP32 to TensorRT GPU inference (120 ms to 3.2 ms), enabling a single GPU server to process 30+ camera feeds simultaneously at 10 fps per camera. The INT8 quantization result — 73% model size reduction and 59% latency reduction at only 1.4% mAP cost — is particularly favorable for edge deployment on resource-constrained hardware such as Raspberry Pi 4 or NVIDIA Jetson Nano.

D. Live Detection Results

The DetectX system was subjected to live operational testing using a standard USB webcam in a realistic indoor environment. A kitchen knife was presented in various orientations and distances over approximately 3 minutes (10:28:49 – 10:31:09) to evaluate detection sensitivity and alarm responsiveness.

Table 5: Live Detection Session Summary Statistics

Metric	**Value**
Session duration	~3 minutes
Total log entries	1,178
Anomaly frames saved	151
Knife detection sequences	5 distinct sequences
Peak knife confidence	0.74
Average knife confidence	~0.47
Alarm events triggered	Multiple (one per sequence)
System crashes	0
Log write failures	0
Frame save failures	0

Table 6: Temporal Analysis of Knife Detection Sequences

Sequence	**Time Range**	**Duration**	**Peak Confidence**	**Alarm**
Sequence 1	10:28:59 – 10:29:18	~19 sec	0.74	Yes
Sequence 2	10:29:25 – 10:29:41	~16 sec	0.57	Yes
Sequence 3	10:29:39 – 10:29:51	~12 sec	0.68	Yes
Sequence 4	10:30:15 – 10:30:28	~13 sec	0.46	Yes
Sequence 5	10:31:06 – 10:31:09	~3 sec	0.70	Yes

E. False Positive Analysis

The live testing session recorded non-knife detections (bed ~320, scissors ~85, remote ~75, laptop ~45, person ~35) that could constitute false positives in an unfiltered system. Critically, none of these triggered alarms because only objects in the DANGER_OBJECTS whitelist (knife) satisfy the alarm condition. The filtering mechanism effectively prevents non-threat detections from generating false alarms, even when the underlying model produces incorrect class predictions for background objects.

F. Feature Performance Analysis

Table 7: Feature Performance Metrics

Module	Metric	Performance
Detection	Accuracy	94.2%
Tracking	ID Consistency	94.0%
Zone Analytics	Accuracy	98.0%
Alerts	Response Time	<100 ms
Database	Write Success	99.9%

G. Cost and Time Efficiency

Building a comparable system from scratch was estimated to require roughly $28,000 and 31 days of engineering effort. Extending YOLO26 reduced this to approximately $20,000 and 20 days. DetectX required only $2,000 and 2 days — representing cost reductions of over 93% and time savings of approximately 94% relative to ground-up development. Over a 24-hour stress test, the system processed close to 8 million frames without crash or observable memory leak, maintaining 100% uptime throughout.

VI. DISCUSSION

A. Significance of the YOLO26 Upgrade

The empirical comparison consistently demonstrates the superiority of YOLO26 for surveillance applications. The 49.2% relative improvement in mAP@0.5:0.95 and the 28.2% improvement in recall are particularly meaningful in threat detection, where recall — the fraction of genuine threats detected — is the more critical metric. A missed detection (a genuine knife unidentified) is operationally more costly than a false positive alarm.

B. Effectiveness of the Persistence Timer

The 2-second persistence timer proved highly effective. Analysis of the anomaly log reveals numerous low-confidence knife detections (confidence 0.25–0.28) appearing as isolated single-frame events between the five main detection sequences — likely from momentary visual similarity or motion blur. These transient detections would each have triggered a false alarm without the persistence filter. By requiring approximately 16–18 consecutive frames of sustained detection, the filter dramatically reduces false alarm rates while imposing only a modest 2-second delay on genuine alarms — a favorable trade-off in surveillance contexts where false alarms erode operator trust.

C. Domain Adaptation as Primary Performance Driver

The central finding of this work is that domain-specific fine-tuning — not architectural novelty — is the primary driver of reliable real-world detection performance. The jump from YOLO26m COCO-pretrained (61.8% knife AP@0.5) to fine-tuned YOLO26m (93.5% knife AP@0.5) demonstrates that the distribution gap between general-

purpose benchmark training data and operational surveillance environments is the dominant performance bottleneck. Investment in domain-specific dataset construction and model fine-tuning yields disproportionately large performance returns compared to architectural innovation alone.

VII. CONCLUSION AND FUTURE WORK

This study demonstrates that a complete, production-ready real-time knife threat detection system can be constructed through systematic integration of YOLO26-based detection with practical engineering of alerting, evidence preservation, and event logging subsystems — at 93% lower cost and 94% less time than ground-up development.

The fine-tuned YOLO26m achieves 87.5% mAP@0.5:0.95 and 93.5% mAP@0.5 on knife detection — a 57.1% relative improvement over the COCO baseline. The temporal persistence filter effectively suppresses false alarms from transient misdetections, and comprehensive deployment benchmarks demonstrate operation across hardware from cloud GPU servers (3.2 ms, 312 fps) to embedded edge devices (45 ms INT8, 3.8 MB). Live testing validated all subsystems with zero failures across 1,178 log entries and 151 saved anomaly frames.

Future work encompasses: (1) multi-threat category extension to firearms and fire/smoke detection; (2) validation on embedded platforms (Jetson Nano, Raspberry Pi 4); (3) transformer-based hybrid detection architectures (DETR, Deformable DETR) for improved small and occluded object detection; (4) federated learning protocols enabling continuous improvement from distributed deployments without sharing raw video data; (5) Grad-CAM explainability integration into anomaly frame archival for interpretable evidence records; and (6) cloud notification integration for remote monitoring and push alerts to security personnel.

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